

$$1) \quad T(z) = \frac{Y(z)}{X(z)}$$

$$a) \quad y(n) = x(n-3) + x(n-2) + x(n-1) + x(n)$$

$$Y(z) = X(z) \cdot z^{-3} + X(z) \cdot z^{-2} + X(z) \cdot z^{-1} + X(z) \cdot z^0$$

$$Y(z) = X(z) \cdot (z^{-3} + z^{-2} + z^{-1} + 1)$$

$$\rightarrow T(z) = \frac{Y(z)}{X(z)} = z^{-3} + z^{-2} + z^{-1} + 1$$

$$b) \quad y(n) = x(n-4) + x(n-3) + x(n-2) + x(n-1) + x(n)$$

$$Y(z) = X(z) \cdot z^{-4} + X(z) \cdot z^{-3} + X(z) \cdot z^{-2} + X(z) \cdot z^{-1} + X(z)$$

$$Y(z) = X(z) \cdot (z^{-4} + z^{-3} + z^{-2} + z^{-1} + 1)$$

$$\rightarrow T(z) = z^{-4} + z^{-3} + z^{-2} + z^{-1} + 1$$

$$c) \quad y(n) = x(n) - x(n-1)$$

$$Y(z) = X(z) \cdot z^0 - X(z) \cdot z^{-1}$$

$$Y(z) = X(z) \cdot (1 - z^{-1})$$

$$\rightarrow T(z) = 1 - z^{-1}$$

$$d) \quad y(n) = x(n) - x(n-2)$$

$$Y(z) = X(z) \cdot z^0 - X(z) \cdot z^{-2}$$

$$Y(z) = X(z) \cdot (1 - z^{-2})$$

$$\rightarrow T(z) = 1 - z^{-2}$$